Лабораторна робота №8

Семенюка Владислава Григоровича

Група ІПЗ-21

Варіант 7

Код консольної програми

```
using Lab8Models;

class Program
{
    static void Main()
    {
        const string divider =
"----------------------------------------------
-----------------------------------";
        const int variant = 7;

        Console.WriteLine(divider);
        Console.WriteLine("Fullname: Vlad
Semeniuk");
        Console.WriteLine("Group: IPZ-21");
        Console.WriteLine("Variant: 7");
        Console.WriteLine(divider);

        Book book = new Book(8 + variant % 8,
$"Book-{variant}", 180 + 5 * variant);
        Magazine magazine = new Magazine(variant,
$"Magazine-{variant}", 40 + 2 * variant);

        Console.WriteLine("Get Base Info");
        Console.WriteLine(book.GetBaseInfo());
        Console.WriteLine(magazine.GetBaseInfo());
        Console.WriteLine(divider);

        Console.WriteLine("Get Info");
```

```
        Console.WriteLine(book.GetInfo());
        Console.WriteLine(magazine.GetInfo());
        Console.WriteLine(divider);

        Console.WriteLine("Call special methods");
        Console.WriteLine($"Average page per
chapter:
{book.CalculateAveragePagesPerChapter():F2}");
        Console.WriteLine($"Issue label:
{magazine.CreateIssueLabel()}");
        Console.WriteLine(divider);

        Console.WriteLine("Try to call special
method with base type of an inherited class\n");
        Publication secondMagazine = new
Magazine(variant, $"Magazine-{variant}", 40 + 2 *
variant);
        // secondMagazine.GetInfo(); it is an error

        ///0>Program.cs(37,24): Error CS1061 :
'Publication' does not contain a definition for
        /// 'GetInfo' and no accessible extension
method 'GetInfo' accepting a first argument of
        /// type 'Publication' could be found (are
you missing a using directive or an assembly
reference?)
        ///0>------- Finished building project:
Lab8App. Succeeded: False. Errors: 1. Warnings: 0

        Console.WriteLine("Explanation: We can't
call method of an inherited class if we initialize
its object with a base type.");
        Console.WriteLine("We could only use method
and properties from the base class");
```

```
        Console.WriteLine("Hierarchy of my
classes");
        Console.WriteLine("--Object\n
|--Publication\n     |--Magazine\n     |--Book");
        Console.WriteLine(divider);

        Console.WriteLine("Individual task A:\n");
        Fiction fiction = new Fiction(title: "It
ends with us", genre: "Romance", chapterCount: 44,
pageCount:223);

Console.WriteLine(fiction.GetDetailedInfo());
        Console.WriteLine($"Is suitable for
children: {fiction.IsSuitableForChildren()}");

        Console.WriteLine(divider);


    }
}
```

```
namespace Lab8Models;

public class Book : Publication
{
    public int ChapterCount { get; }
```

```
    public Book(int chapterCount, string title, int
pageCount)
        : base(title, pageCount)
    {
        ChapterCount = chapterCount;
    }


    public double CalculateAveragePagesPerChapter()
    {
        return (double)this.PageCount /
this.ChapterCount;
    }

    public string GetInfo()
    {
        return $"{base.GetCommonDescription()}, " +
               $"chapter count: {ChapterCount}";
    }
}
```

```
namespace Lab8Models;

public class Fiction : Book
{
    public string Genre { get; }

    public Fiction(string genre, int chapterCount,
string title, int pageCount)
        : base(title: title, pageCount: pageCount,
chapterCount: chapterCount)
    {
        Genre = genre;
    }

    public bool IsSuitableForChildren()
```

```
    {
        string normalized =
Genre.ToLowerInvariant();
        return !(normalized.Contains("horror") ||
normalized.Contains("thriller") ||
normalized.Contains("crime"));
    }

    public string GetDetailedInfo()
    {
        return base.GetInfo() + $", genre:
{Genre}";
    }

}
```

```
namespace Lab8Models;

public sealed class Magazine : Publication
{
    public int IssueNumber { get; }

    public Magazine(int issueNumber, string title,
int pageCount)
        : base(title, pageCount)
    {
        IssueNumber = issueNumber;
    }


    public string CreateIssueLabel()
    {
        return $"{this.Title}-#{this.IssueNumber}";
    }

    public string GetInfo()
```

```
    {
        return $"{base.GetCommonDescription()}, " +
               $"issueNumber: {IssueNumber}";
    }
}
```

```
namespace Lab8Models;

public class Publication
{
    public string Title { get; }
    public int PageCount { get; }

    public Publication(string title, int pageCount)
    {
        Title = title;
        PageCount = pageCount;
    }

    public string GetBaseInfo()
    {
        return $"Base Info:
{GetCommonDescription()}";
    }

    protected string GetCommonDescription()
    {
        return $"title: {Title}, pageCount:
{PageCount}";
    }
}
```

ШІ використовувався тільки для ресьорчу.